Paper	ENCN342 – Hydraulics and Applied Hydrology
Topic	Steady Open Channel Flow
Module	5 – Gradually Varied Flow - addition

Summary

This is an addition to the module that presents the concept of gradually varied flow. The analysis of free surface profiles along entire reaches is presented. This analysis is strongly dependent on the classification of channel slopes. The focus here is on the quantitative analysis of water surface profiles.

Prerequisites

Module 4

Objectives – the section highlighted in bold is discussed here in more detail

At the conclusion of this module you should:

1. Be able to define gradually varied flow and be able to derive the differential equation that describes the variation of flow depth along a channel.

2. Understand and be able to explain the key principles upon which gradually varied flow analysis is based.

3. Be able to classify channel slopes.

4. Be able to deduce the standard free surface profiles for mild and steep sloped channels.

5. Be able to use the standard free surface profiles to analyse the free surface profile along a channel reach that may include a number of flow controls.

6. Understand how the discharge problem is handled.

7. **Be able to implement either of the quantitative methods for computing free surface profiles.**

Key Points

1. Gradually varied flow

See Module 5 Summary.

2. Key principles

See Module 5 Summary.

3. Governing equations

See Module 5 Summary.

4. Classification of channel slopes

See Module 5 Summary.

5. Longitudinal surface profiles

See Module 5 Summary.

6. Qualitative reach analysis

See Module 5 Summary.

7. The discharge problem

See Module 5 Summary.

8. Quantitative profile calculation – addition

$$\frac{dy}{dx} = f(y) \tag{4}$$

This equation is generally integrated numerically. The channel is discretised into a series of nodes labelled by an index *i*.

Finite difference form of equation (4)

$$\frac{y_{i+1} - y_i}{x_{i+1} - x_i} = f(y_i, y_{i+1}) \tag{5}$$

8.1 Numerical Integration Method

$$\frac{dy}{dx} = \frac{S_0 - \dfrac{Q^2 n^2}{A^2 R^{4/3}}}{1 - \dfrac{Q^2 B}{g A^3}} \tag{6}$$

Equation (6) is a nonlinear equation as *A*, *B* and *R* on the right-hand side of the equation are dependent on *y* for a given section. For wide rectangular sections equation (6) is written as

$$\frac{dy}{dx} = \frac{S_0 - \dfrac{q^2 n^2}{y^{10/3}}}{1 - \dfrac{q^2}{g y^3}} \tag{7}$$

$$y_{i+1} = y_i + dy \tag{8}$$

or

$$y_{i+1} = y_i + \frac{dy}{dx} dx \tag{9}$$

If the distance *dx* is chosen small enough, then *dy*/*dx* can be satisfactorily assumed to vary linearly over the length increment *dx*. Therefore,

$$y_{i+1} = y_i + \frac{y'_{i+1} + y'_i}{2} dx \tag{10}$$

where $y'_i = \dfrac{dy_i}{dx}$ and $y'_{i+1} = \dfrac{dy_{i+1}}{dx}$. Equations (7) and (10) can be solved numerically to determine the water surface profile as follows:

1. Start from the given depth y_i, and calculate y'_i using equation (7).
2. As a first trial and to initiate the solution, let $y'_{i+1} = y'_i$.
3. Using equation (10), calculate the value of y_{i+1}; this is an approximate value.
4. Using equation (7), calculate y'_{i+1}. Substitute the new value of y'_{i+1} into equation (10) to get y_{i+1}.
5. If the values of y_{i+1} as obtained from steps 3 and 4 are not close to each other, repeat steps 3 and 4.
6. The correct solution for y_{i+1} is obtained when its values as calculated from steps 3 and 4 are the same (or within a specified tolerance). Proceed to the next section located after a distance *dx*. The last computed y_{i+1} now becomes y_i. The above steps are repeated until the profile is completed.

8.2 Direct Step Method

The direct step method is applicable to any uniform prismatic channel and can be employed using either the Manning or Chezy formula.

$$S_0 dx + y_1 + \frac{u_1^2}{2g} = y_2 + \frac{u_2^2}{2g} + S_f dx \tag{11}$$

Hence,

$$(S_0 - S_f)dx = \left(y_2 + \frac{u_2^2}{2g} \right) - \left(y_1 + \frac{u_1^2}{2g} \right) = E_2 - E_1 \tag{12}$$

S_f, may vary from one section to the other. Because we are calculating the length of a water surface profile between two sections, S_f in equation (12) should be replaced with the average value $\bar{S}_f$, where

$$\bar{S}_f = \frac{S_{f1} + S_{f2}}{2} \tag{13}$$

where the individual point estimates of S_f can be computed from e.g. using Manning, and hence $S_f = \frac{Q^2 n^2}{A^2 R^{4/3}}$.

Then equation (12) can be written as

$$dx = \frac{E_2 - E_1}{S_0 - \bar{S}_f} = \frac{\Delta E}{S_0 - \bar{S}_f} \tag{14}$$

Direction of the computation:

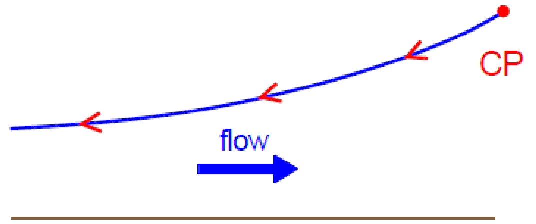

Figure 1. Computation begins at downstream control point and proceeds upstream for subcritical conditions.

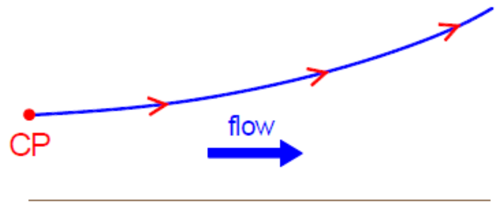

Figure 2. Computation begins at upstream control point and proceeds downstream for supercritical conditions.

To calculate the distance between any two given depths, one may divide the reach under consideration into several subreaches. The lengths of profiles are calculated for subreaches and then summed to give the required length. Since the direct step method requires many repetitive calculations, it is best employed using a computer spreadsheet or e.g. a Python script.

8.3 Standard Step Method

The standard step method is completely general and can be employed in any steady flow situation regardless of the cross-sectional shape or channel conditions. The execution of the standard step method requires that surveyed channel cross-sections be available at known locations along the stream.
Case discussed here: the flow is subcritical and all conditions are known at section 1. Then the conditions (depth and velocity) at the upstream section 2 are to be determined.

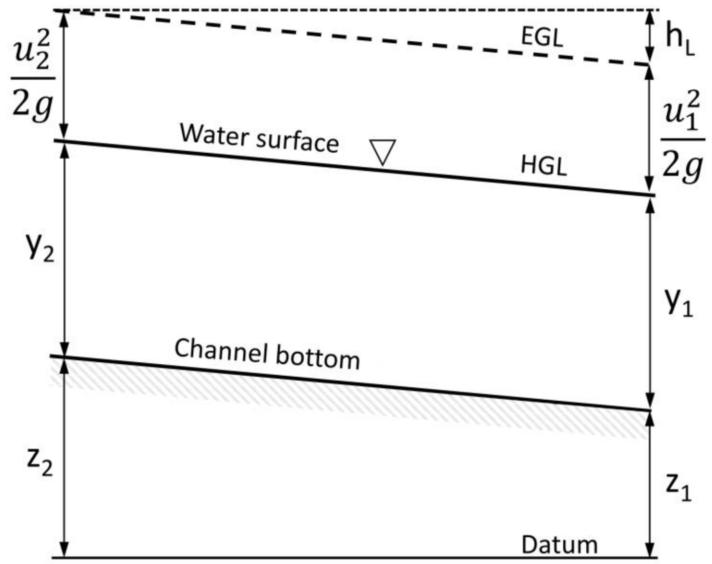

Figure 3. Notation diagram for the standard step method.

In the execution of the standard step method, the stage ($WS = z + y$) is usually employed from the static energy term. Bernoulli equation:

$$WS_1 + \frac{u_1^2}{2g} + h_L = WS_2 + \frac{u_2^2}{2g} \tag{15}$$

Where WS = water surface elevation (stage) and h_L = total head loss between sections 1 and 2.
Steps:

1. A value of stage is assumed at section 2 (WS_2) and all hydraulic variables are computed for this section from the assumed stage.
2. Head loss computed from the equation (15), termed the ‘observed head loss’.
3. ‘Calculated head loss’, determined from a resistance equation (usually Manning) and any minor losses can be added in to result in the calculated head loss for the channel reach:

$$h_L = \overline{S}_f L + C\left|\frac{u_2^2}{2g} - \frac{u_1^2}{2g}\right| \tag{16}$$

where $\overline{S}_f$ = average friction slope between the two sections (! friction slope, not total !), L = length of the channel between the two sections, and C = expansion or contraction coefficient, whichever is applicable.

4. The two estimates of the head loss are then compared and if they do not match within a specified tolerance, a new stage value is assumed at section 2 and the computations are repeated.
5. The next estimate of the stage at section 2 can be obtained from the following:

$$WS_1 + \frac{u_1^2}{2g} + h_{L,cal} = WS_2 + \frac{u_2^2}{2g} \tag{17}$$

where $h_{L,cal}$ is the head loss calculated from the friction slope and minor losses in the previous step and $\frac{u_2^2}{2g}$ is the velocity head at section 2 obtained from the previous step.
6. Then, with all conditions at section 1 known, equation (17) is solved for WS2. This would be the necessary stage at section 2 if the head loss was really equal to the calculated value as opposed to the observed value. The new assumed stage is then estimated as the average of the previous value and the value computed from equation (17).

As in the direct step method, because the procedure consists of many repetitive trials and computations, the solution is greatly facilitated by the use of spreadsheets, or can e.g. be coded in a Python script. This method is also commonly implemented in computer software packages such as HEC-RAS.

8.4 Practicalities

For the direct and standard step methods the computation is performed upstream and downstream of each control in the channel. The control will provide the boundary condition (i.e. the depth at a particular location). The upstream computation will be for subcritical profiles and the downstream computation for supercritical profiles.

In regions where conflicting profiles exist the numerical method can include the computation of conjugate depths that will enable the location of hydraulic jumps to be determined.

Worked Examples

Question 1

A wide rectangular channel with a slope of 0.0006 conveys a specific discharge of 3.5 m^2/s. A gate is located midway in the channel length. The depth of water at the 'vena contracta', immediately downstream of the gate, is 0.25 m. Draw the water surface profile and estimate the length of the rising curve formed after the gate. Take the Manning roughness coefficient as 0.02 (metric).

Solution

To draw the water surface profile, the critical and normal depths are calculated first:

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{3.5^2}{9.81}} = 1.08m$$

for wide rectangular channel sections, the normal depth can be calculated as

$$y_0 = \left(\frac{q^2 n^2}{S_0}\right)^{0.3} = \left(\frac{3.5^2 \times 0.02^2}{0.0006}\right)^{0.3} = 1.88m$$

The water surface profile is drawn as shown in Figure 9. The normal depth constitutes the high stage depth of the jump. The low stage depth can be calculated as

$$y_1 = \frac{y_0}{2}\left(\sqrt{1+8\left(\frac{y_c}{y_0}\right)^3} - 1\right) = \frac{1.88}{2}\left(\sqrt{1+8\left(\frac{1.08}{1.88}\right)^3} - 1\right) = 0.55m$$

Then an *M3* curve will form through which the depth of flow increases from 0.25 m to 0.55 m as shown in Figure 4.

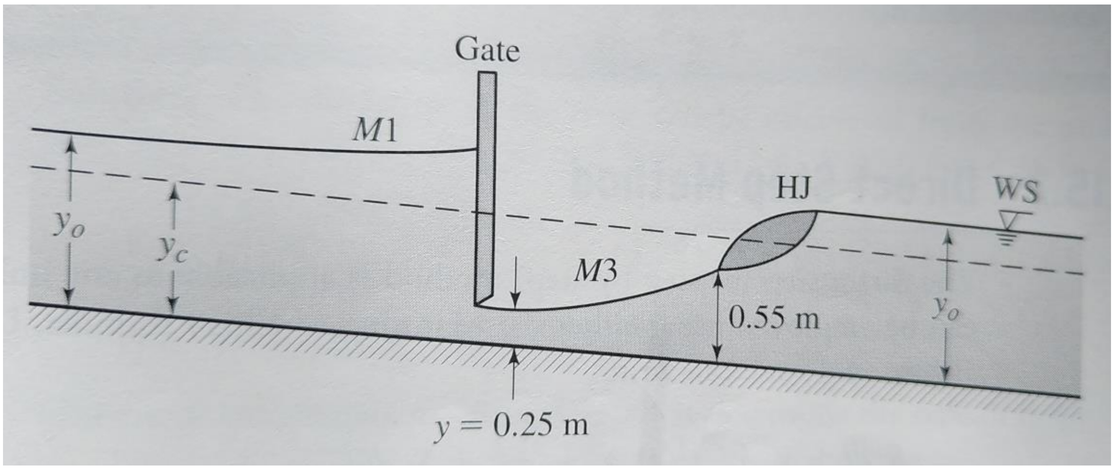

Figure 4. Water surface profile.

To estimate the length of the *M3* curve, a stepwise procedure is followed. Let y_1 = 0.25 m and take dx_1 = 40 m, for example. Then

$$y_i' = \frac{0.0006 - \dfrac{3.5^2 \times 0.02^2}{0.25^{10/3}}}{1 - \dfrac{3.5^2}{9.81 \times 0.25^3}} = \frac{-0.497}{-78.92} = 0.0063$$

Assuming $y_{i+1}' = y_i'$

$$y_{i+1} = 0.25 + 0.0063 \times 40 = 0.502m$$

$$y_{i+1}' = \frac{0.0006 - \dfrac{3.5^2 \times 0.02^2}{0.502^{10/3}}}{1 - \dfrac{3.5^2}{9.81 \times 0.502^3}} = \frac{-0.0481}{-8.87} = 0.0054$$

Then

$$y_{i+1} = 0.25 + \left(\frac{0.0054 + 0.0063}{2}\right) \times 40 = 0.484m$$

Using the new value of y_{i+1}, y_{i+1}' is evaluated again,

$$y_{i+1}' = \frac{0.0006 - \dfrac{3.5^2 \times 0.02^2}{0.484^{10/3}}}{1 - \dfrac{3.5^2}{9.81 \times 0.484^3}} = \frac{-0.0544}{-10.014} = 0.0054$$

Then the last value of y_{i+1} is correct. Now let $y_i = y_{i+1} = 0.484$ m and take dx_2 as 15 m only, as we need to reach the depth of 0.55 m. Then

$$y_{i+1} = 0.484 + 0.0054 \times 15 = 0.565m$$

$$y_{i+1}' = \frac{0.0006 - \dfrac{3.5^2 \times 0.02^2}{0.565^{10/3}}}{1 - \dfrac{3.5^2}{9.81 \times 0.565^3}} = \frac{-0.032}{-5.923} = 0.0054$$

Hence the calculated value for $y_{i+1} = 0.565$ is correct.

One may get the exact length to the depth of 0.55 m by interpolation. The last 15 m on the *M3* curve gave an increase in the flow depth by 0.0054 x 15 = 0.081 m. However, one needs to get the curve length which gives an increase in the depth of 0.55 – 0.484 = 0.066 m. Then

$$dx_2 = 15 \times \frac{0.066}{0.081} = 12.2m$$

Then the total length of the *M3* curve before the hydraulic jump = $dx_1 + dx_2$ = 40 + 12.2 = 52.2 m.

Question 2
Solve the problem given in Question 1 using the direct step method.

Solution
We consider the two depths corresponding to the beginning and end of the *M3* profile as 0.25 m and 0.55 m, respectively. We also consider two other intermediate depths. Then y_1 = 0.25 m, y_2 = 0.35 m, y_3 = 0.45 m, and y_4 = 0.55 m.

For depth y_1 = 0.25 m:
$u_1 = q/y_1 = 3.5/0.25 = 14.00$ m/s
$E_1 = y_1 + u_1^2/(2g) = 0.25 + 14.00^2/(2 * 9.81) = 10.24$ m
$S_{f1} = n^2 * u_1^2/R_1^{(4/3)} = 0.02^2 * 14.00^2/0.25^{(4/3)} = 0.4978$

For depth y_1 = 0.25 m:
$u_2 = q/y_2 = 3.5/0.35 = 10.00$ m/s
$E_2 = y_2 + u_2^2/(2g) = 0.35 + 10.00^2/(2 * 9.81) = 5.45$ m
$S_{f2} = n^2 * u_2^2/R_2^{(4/3)} = 0.02^2 * 10.00^2/0.35^{(4/3)} = 0.1622$

$\Delta E = E_2 - E_1 = 5.45 - 10.24 = -4.79$ m
$<S_f> = (S_{f1} + S_{f2})/2 = (0.4978 + 0.1622)/2 = 0.3300$
$S_0 - <S_f> = 0.0006 - 0.3300 = -0.3294$
$\Delta x = \Delta E/(S_0 - <S_f>) = -4.79/-0.3294 = 14.54$ m

Note: < > is used to denote the average value.

Calculations can be organised conveniently in a spreadsheet (the results differ slightly, as less decimal places are carried through in the hand calculations above).

y	u=q/y	E=y+u^2/(2g)	ΔE=E2-E1	Sf=n^2*u^2/R^(4/3)	<Sf>=(Sf,i+Sf,i+1)/2	S0-<Sf>	dx=dE/(S0-<Sf>)
[m]	[m/s]	[m]	[m]				[m]
0.25	14	10.24		0.49780897			
			-4.79		0.329989251	-0.329389251	14.55
0.35	10	5.45		0.162169533			
			-1.91		0.11617	-0.11557	16.56
0.45	7.778	3.53		0.070170467			
			-0.92		0.053058366	-0.052458366	17.52
0.55	6.364	2.61		0.035946266			
							48.63

(wide rectangular channel, so R=y)

The resulting length of the M3 profile is 48.63 m, which is close to the calculated length in Question 1 via the numerical integration method.

Question 3

Water flows in a 5 m wide rectangular channel made from unfinished concrete with n = 0.015 (metric). The channel contains a long reach on which S_0 is constant at S_0 = 0.020. At section 1, flow is at depth y_1 = 1.5 m, with velocity u_1 = 4.0 m/s. Estimate the channel location y_2 where the flow reaches a depth of 0.9 m.

Solution

The location where y_2 = 0.9 m may be upstream or downstream from section 1, depending on y_1, y_c and y_0. The flow rate is

$$Q = u_1 b y_1 = 4 \times 5 \times 1.5 = 30 \text{ m}^3/\text{s}$$

At critical condition, u = $u_c = \sqrt{g u_c}$, so

$$Q = u_1 b y_1 = u_c b y_c = \sqrt{g u_c} b y_c = b\sqrt{g}\, y_c^{3/2}$$

Thus

$$y_c = \left(\frac{Q}{b\sqrt{g}}\right)^{2/3} = \left(\frac{30}{5\times\sqrt{9.81}}\right)^{2/3} = 1.54 \text{ m}$$

Since $y < y_c$, the flow is supercritical.

$$Fr_1 = \frac{u_1}{\sqrt{g y_1}} = \frac{4}{\sqrt{9.81\times 1.5}} = 1.04 > 1$$

The uniform flow depth at this slope, roughness and discharge can be determined from (using $R = A/P = (by/(b + 2y))$

$$Q = \frac{1}{n} R^{2/3} S^{1/2} A = \frac{1}{n} R^{2/3} S^{1/2} A = \frac{1}{n}\left(\frac{y_0}{1+2y_0/b}\right)^{2/3} S^{1/2} b y_0 = 47.14 y_0 \left(\frac{y_0}{1+2y_0/b}\right)^{2/3}$$

By trial and error or using e.g. the goal seek function in Excel we find y_0 = 0.8823 m. Thus $y_c > y > y_0$, so the flow follows an S2 curve. We expect the depth to decrease in the direction of flow and y should approach y_0 = 0.88 asymptotically.

Use the direct step method equation

$$\Delta x = \frac{\Delta E}{S - \bar{S}_f}$$

to determine the channel reach Δx.

First express E as a function of y and evaluate $\bar{S}_f$.

Using $u = Q/(by)$

$$E = \frac{u^2}{2g} + y = \frac{Q^2}{2gb^2y^2}$$

and

$$\bar{S}_f = \frac{n^2 \bar{u}^2}{\bar{R}^{4/3}}$$

where $\bar{u} = \frac{1}{2}\left(u_{i+1} + u_i\right)$ and $\bar{R} = \frac{1}{2}\left(R_{i+1} + R_i\right)$

The equations for calculating quantities in terms of y are

$$u=\frac{Q}{by}=\frac{u_1 b y_1}{by}=\frac{u_1 y_1}{y}=\frac{4\times 1.5}{y}=\frac{6.000}{y}$$

$$E=\frac{Q^2}{2gb^2y^2}+y=\frac{30^2}{2\times 9.81\times 5^2 y^2}+y=\frac{1.835}{y^2}+y$$

$$R=\frac{A}{P}=\frac{by}{b+2y}=\frac{y}{1+2y/b}=\frac{y}{1+2y/5}=\frac{y}{1+0.400\times y}$$

Calculations (spreadsheet solution shown):

y	**u**	**E**	**R**	**<R>**	**<u>**	**<Sf>**	**delta x**	**x**
[m]	[m/s]	[m]	[m]	[m]	[m/s]	[-]	[m]	[m]
1.5	4.000	2.316	0.9375					0
				0.9175	4.143	0.004332	1.319	
1.4	4.286	2.336	0.8974					1.32
				0.8763	4.451	0.005314	3.376	
1.3	4.615	2.386	0.8553					4.69
				0.8330	4.808	0.006635	6.622	
1.2	5.000	2.474	0.8108					11.32
				0.7873	5.227	0.008456	12.320	
1.1	5.455	2.617	0.7639					23.64
				0.7391	5.727	0.011045	24.395	
1.0	6.000	2.835	0.7143					48.03
				0.6880	6.333	0.014858	64.268	
0.9	6.667	3.165	0.6618					112.30

Graphical representation: